# KEY2USB KIT

Use your Commodore 64's own keyboard as a USB keyboard for VICE — no soldering into the machine, no case modification.

**Thank you for your order!** Everything you need is in this box: the PCB, all components, and a 3D-printed case. **The two programmed chips — the 28C64 EEPROM (U1) and the ATmega328 (U5) — arrive pre-programmed and ready to run**, so there is nothing to flash. All you need to supply is a soldering iron, solder, and side cutters. If anything is missing or damaged, email me and I'll make it right.

**⚠ SET VICE TO POSITIONAL**

**Settings → Keyboard → Keyboard Mapping → Positional** — not Symbolic. This is the single most common first-time mistake; if your keys come out wrong, check it before anything else. KEY2USB reports keys by *physical position*, never as characters. Symbolic layers an unwanted PC-layout translation on top of that and garbles shifted keys — SHIFT + 2 arrives as `@` instead of `"`.

## WHAT'S IN THE KIT

| QTY | REF | PART |
|---|---|---|
| 1 | — | KEY2USB PCB + 3D-printed case |
| 1 | **U1** | 28C64 EEPROM + socket — **pre-programmed** |
| 1 | **U5** | ATmega328 MCU + socket — **pre-programmed** |
| 3 | **U2–U4** | 74LS32 (DIP-14), 74LS273 (DIP-20), 74LS74 (DIP-14) glue logic |
| 3 | **Y1, C5, C6** | 16 MHz crystal + 2× 20pF load caps |
| 4 | **C1–C4** | 100nF ceramic — IC bypass |
| 2 | **D1, D2** | 1N5227B 3.6V Zener — *polarized* |
| 3 | **R1–R3** | 2× 68Ω USB series, 1× 1.5kΩ D– pull-up |
| 2 | **J1, J3** | 3-pin MODE header + jumper cap, USB Type B connector |

## ASSEMBLY

All through-hole. Build shortest-to-tallest so parts sit flat. The two programmed chips get **sockets**; the three 74LS logic ICs solder **directly** down.

1. Resistors `R1–R3`, then ceramic capacitors `C1–C6` — neither is polarized.
2. Zener diodes `D1 D2` — **polarized.** The band on the body must match the banded end of the silkscreen outline.
3. The two DIP-28 sockets at `U1` and `U5`, notch matching the silkscreen. Leave them empty for now.
4. `U2 U3 U4` soldered straight down — **check each notch against the silkscreen first**; these are not socketed and are unpleasant to remove.
5. Crystal `Y1`, MODE header `J1`, then the USB Type B connector `J3` — seat J3 fully flat; it takes the mechanical stress of the cable.
6. Trim all leads flush, then inspect for solder bridges — especially along the edge fingers and between adjacent IC pins.
7. Press the labeled `U1` and `U5` chips into their sockets last, notch matching, with no bent-under pins.

**Before closing the case:** fit the jumper cap on `J1` in the **C64** position (the GND side, marked on the silkscreen) — right for every machine listed below. C128 owners may choose the **C128** position instead; see Compatibility.

## COMPATIBILITY

| | |
|---|---|
| **Commodore 64 / 64C** | Yes — primary target |
| **Commodore 128 / 128D** | Yes — C64 mode, or native C128 mode |
| **Ultimate 64** | Yes — verified on hardware |

**C128, J1 on C64:** the cartridge asserts `/EXROM`, so the C128 comes up in C64 mode on its own (if one does not autostart, power on holding `C=`). Every key it shares with the C64 works; its extra keys are not reported.

**C128, J1 on C128:** the cartridge runs in native C128 mode and sends the whole keyboard — keypad, ESC, TAB, ALT, HELP, LINE FEED, NO SCROLL, cursor pad, CAPS LOCK, 40/80 — to VICE's `x128` (Positional there too). To type on the keypad, set Joystick Port #2 to **None**. Use this position on a C128 only.

## FIRST POWER-ON

1. With the C64 **powered off**, insert KEY2USB into the expansion port.
2. Connect the USB cable to your PC. It enumerates as a standard HID keyboard — no drivers needed.
3. Power on the C64. The `KEY2USB` splash screen appears; the cartridge has taken over the machine and is scanning the keyboard.
4. In VICE, set **Settings → Keyboard → Keyboard Mapping → Positional**. This step is mandatory.
5. Type on the C64. Keystrokes arrive at the PC.

While KEY2USB runs, the C64 is a dedicated keyboard; power off and remove the cartridge to use it normally. RUN/STOP + RESTORE works in VICE (RESTORE arrives as Page Up).

## JOYSTICKS IN VICE

Both ports arrive as keypad keystrokes. First tick **Settings → Joystick → Allow keyset joysticks** (in each VICE emulator you use).

**Port 2 — Settings → Joystick → Joystick Port #2 → Numpad**. VICE's built-in Numpad device matches the cartridge exactly.

**Port 1** — needs a keyset, since Numpad can only serve one port. Set **Joystick Port #1 → Keyset A**, then **Configure Keyset A**:

| DIRECTION | KEY |
|---|---|
| **North** | Keypad **–** |
| **South** | Keypad **+** |
| **West** | Keypad **/** |
| **East** | Keypad ***** |
| **Fire** | Keypad **enter** |

**Worth knowing:** a held joystick direction grounds a line in the keyboard matrix, so it phantom-presses real keys — exactly as on a bare C64. Port 1 is hit hardest, grounding a whole matrix row (RETURN, INST/DEL, CLR/HOME), so expect stray keystrokes while a port-1 direction is held. **Port 2 is the one to use.**

## TROUBLESHOOTING

| SYMPTOM | CHECK |
|---|---|
| **Wrong characters** | VICE keyboard mapping is set to **Positional**, not Symbolic. |
| **No splash screen** | Cartridge fully seated, C64 powered off when inserted. Then re-check IC orientation and bridged joints near U1 and the edge fingers. |
| **No USB device** | The C64 must be powered on — the cartridge is bus-powered. Then check Y1 and its 20pF caps, R1–R3, and the D1/D2 band orientation. |
| **Joystick does nothing** | **Allow keyset joysticks** ticked; port 2 **Numpad**; port 1 **Keyset A**, configured as above. |
| **Shift stays on** | SHIFT LOCK is down — it is wired as LEFT SHIFT. |

Repository `Software/`: `key2usbmon` (macOS) shows what the cartridge sends, without VICE; `key2usbtest.prg` shows what VICE receives.

**Schematic, firmware source & full documentation:** github.com/acwright/KEY2USB
**Store:** tindie.com/stores/acwrightdesign · **Support:** acwrightdesign@gmail.com

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Stuck on a build? Please reach out before leaving a review — I'm happy to help.

**— A.C. Wright**